

- 3** An object is projected with velocity 40 m s^{-1} at an angle of 60° to the horizontal. Air resistance is negligible.

What is the speed of the object after 5.0 s ?

- A** 14 m s^{-1} **B** 25 m s^{-1} **C** 35 m s^{-1} **D** 45 m s^{-1}

Ans: B

The horizontal velocity is constant at $v_x = u_x = 40 \cos 60^\circ = 20.0 \text{ m s}^{-1}$.

The initial vertical velocity is $u_y = 40 \sin 60^\circ = 34.6 \text{ m s}^{-1}$.

After 5.0 s, $v_y = u_y + a_y t = 34.6 + (-9.81)(5.0) = -14.4 \text{ m s}^{-1}$.

Hence, after 5.0 s, $v = \sqrt{v_x^2 + v_y^2} = 25 \text{ m s}^{-1}$.

- 4 A block with a pendulum bob hanging lies on a rough surface. The total mass of the block and